Long-run variance estimation

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HAC

$$\Omega \equiv \mathrm{LRV}(y_t) = \sum_{\ell=-\infty}^{\infty} \Gamma_y(\ell) = 2\pi s_y(0)$$

- Need estimate $\hat{\Omega}$ of LRV Ω for standard errors, conf. intervals, and hypothesis tests.
- A consistent estimator $\hat{\Omega}$ is called **heteroskedasticity and autocorrelation consistent (HAC)**.
- Unfortunately, the most natural-looking estimator

$$\hat{\Omega} \equiv \sum_{\ell=-(T-1)}^{T-1} \hat{\Gamma}_y(\ell)$$

turns out to be inconsistent. Intuitively, we're including many very noisy $\hat{\Gamma}_y(\ell)$ for $\ell \approx T$.

Hansen-Hodrick standard errors

- In some applications (like multi-step forecasting) we may be willing to assume that $\{y_t\}$ is a VMA(q) process with finite q.
- Then $\Omega = \sum_{\ell=-q}^{q} \Gamma_y(\ell)$, which can be estimated consistently by the **Hansen & Hodrick (1980) estimator**

$$\hat{\Omega} \equiv \sum_{\ell=-q}^{q} \hat{\Gamma}_y(\ell).$$

- Serious practical problem: $\hat{\Omega}$ may not be positive definite in finite samples $\Rightarrow$ negative asy. variance estimate for individual parameters. Unclear how to proceed with std. errors, conf. intervals, or tests.

VAR-HAC

- One solution: approximate the DGP with a VAR(p) model.
- Exercise: Suppose true DGP is stationary VAR(p), $A(L)y_t = \nu + u_t$, $u_t \sim WN(0, \Sigma)$. Then $\Omega = A(1)^{-1}\Sigma A(1)^{-1\prime}$.
- Feasible estimation procedure:
  1. Choose p using a model selection method.
  2. Estimate the VAR(p) parameters $\hat{A}_\ell$ and $\hat{\Sigma}$ using least-squares.
  3. Construct

     $$\hat{\Omega} \equiv \left(I_n - \textstyle\sum_{\ell=1}^{p} \hat{A}_\ell\right)^{-1} \hat{\Sigma} \left(I_n - \textstyle\sum_{\ell=1}^{p} \hat{A}_\ell\right)^{-1\prime}.$$

     Always positive semidefinite (p.s.d.).
- Den Haan & Levin (1998) show that if p is chosen by BIC, then $\hat{\Omega}$ is consistent for Ω under regularity conditions (even if y_t is not a finite-dim. VAR). They call this LRV estimator **VAR-HAC**.

Periodogram smoothing

- We have also talked about a second strategy for LRV estimation: periodogram smoothing applied at frequency $\omega = 0$.

- Given weight function $w \colon [-1,1] \to \mathbb{R}_+$ symmetric around 0 and satisfying $w(0) = 1$, and given bandwidth $B_T \in \mathbb{N}$, we estimate

$$\hat{\Omega} \equiv \textstyle\sum_{|\ell| \leq B_T} w_{B_T}(\ell) \hat{\mathcal{I}}(\omega_\ell),$$

  where $\omega_\ell \equiv \frac{2\pi\ell}{T}$, $w_{B_T}(\ell) \propto w(\ell/B_T)$, and $\sum_{|\ell| \leq B_T} w_{B_T}(\ell) = 1$.

- We discussed how periodogram smoothing consistently estimates the spectrum (at any frequency) when $B_T \to \infty$, $B_T/T \to 0$.

- We will now reinterpret this estimator $\hat{\Omega}$. Recall the result

$$\hat{\mathcal{I}}(\omega_\ell) = \textstyle\sum_{m=-(T-1)}^{T-1} e^{-i\omega_\ell m} \hat{\Gamma}_y(m).$$

  We henceforth define $\hat{\mathcal{I}}(0) \equiv 0_{n \times n}$, so the above formula also holds for $\omega_\ell = 0$ (exercise: verify).

Periodogram smoothing (cont.)

- Express $\hat{\Omega}$ in terms of $\hat{\Gamma}_y(\cdot)$:

$$\hat{\Omega} \equiv \sum_{|\ell| \leq B_T} w_{B_T}(\ell) \sum_{m=-(T-1)}^{T-1} e^{-i\omega_\ell m} \hat{\Gamma}_y(m)$$
$$= \sum_{m=-(T-1)}^{T-1} k_{w,B_T}(\tfrac{m}{T}) \hat{\Gamma}_y(m),$$

  where we define the discrete Fourier transform of $w_{B_T}(\cdot)$:

$$k_{w,B_T}(x) \equiv \sum_{|\ell| \leq B_T} w_{B_T}(\ell) e^{-i\omega_\ell Tx} = \sum_{|\ell| \leq B_T} w_{B_T}(\ell) \cos(2\pi \ell x).$$

- Hence, a smoothed periodogram estimator of the LRV corresponds to a weighted sum of the sample ACF at different lags.

Kernel estimators

- The correspondence with periodogram smoothing suggests that we analyze the **kernel HAC estimator**

$$\hat{\Omega} \equiv \sum_{\ell=-(T-1)}^{T-1} k(\tfrac{\ell}{S_T})\hat{\Gamma}_y(\ell),$$

  where the **kernel function** $k\colon \mathbb{R} \to [-1,1]$ satisfies $k(-x) = k(x)$ and $k(0) = 1$.

- $S_T \in \mathbb{N}$ is a **bandwidth**. To avoid confusion with the periodogram smoothing bandwidth B_T, we call the latter the *spectral bandwidth*.

- Intuitively, $\hat{\Omega}$ is like a sample analogue of $\Omega = \sum_{\ell=-\infty}^{\infty} \Gamma_y(\ell)$, except we down-weight the imprecisely estimated $\hat{\Gamma}_y(\ell)$ for large $|\ell|$.

- For $S_T \ll T$, a (time domain) kernel HAC estimator is approx. equal to a (frequency domain) periodogram smoothing LRV estimator with $B_T \propto T/S_T$ and a particular spectral weight fct (Müller, 2014).

- Hence, if true spectrum is flat near 0, kernel HAC should do well.

Kernel estimators: positive semidefiniteness

- When is $\hat{\Omega}$ positive semidefinite (like Ω) in finite samples?
- Suppose there exists a $w(\cdot) \geq 0$ s.t. $k(x) = \int_{-\infty}^{\infty} e^{-i\lambda x} w(\lambda)\, d\lambda$.
- Then for any $v \in \mathbb{R}^n$,

$$
\begin{aligned}
v'\hat{\Omega}v &= \sum_{\ell=-(T-1)}^{T-1} \left(\int_{-\infty}^{\infty} e^{-i\lambda\ell/S_T} w(\lambda)\, d\lambda \right) v'\hat{\Gamma}_y(\ell)v \\
&= \frac{T}{2\pi S_T} \sum_{\ell=-(T-1)}^{T-1} \left(\int_{-\infty}^{\infty} e^{-2\pi i\tau\ell/T} w(\tfrac{2\pi\tau S_T}{T})\, d\tau \right) v'\hat{\Gamma}_y(\ell)v \\
&\propto \int_{-\infty}^{\infty} \underbrace{\left(\sum_{\ell=-(T-1)}^{T-1} e^{-i\omega_\tau \ell} v'\hat{\Gamma}_y(\ell)v \right)}_{\geq 0} \underbrace{w(\omega_\tau S_T)}_{\geq 0}\, d\tau,
\end{aligned}
$$

since the big parenthesis is the periodogram of $v'y_t$ at freq. $\omega_\tau = \frac{2\pi\tau}{T}$.

Kernel estimators: kernels

- It's customary to restrict attention to kernels $k(\cdot)$ that satisfy the sufficient condition for positive semidefiniteness on the previous slide.

- Examples:

1. Bartlett kernel: $k(x) = \max\{1 - |x|, 0\}$. This yields the famous and widely used **Newey & West** (1987) HAC estimator.

2. Parzen kernel: $k(x) = \begin{cases} 1 - 6x^2 + 6|x|^3, & 0 \leq |x| < 1/2, \\ 2(1 - |x|)^3, & 1/2 \leq |x| \leq 1, \\ 0, & |x| > 1. \end{cases}$

3. Quadratic spectral (QS), also known as Epanechnikov:

$$k(x) = \frac{25}{12\pi^2 x^2}\left(\frac{\sin(6\pi x/5)}{6\pi x/5} - \cos(6\pi x/5)\right) \text{ for } x \in \mathbb{R}\backslash\{0\}, \quad k(0) = 1.$$

- The intuitively pleasing rectangular kernel $k(x) = \mathbb{1}(|x| \leq 1)$ does not guarantee a p.s.d. $\hat{\Omega}$ (Hansen-Hodrick).

Kernel estimators: kernels (cont.)

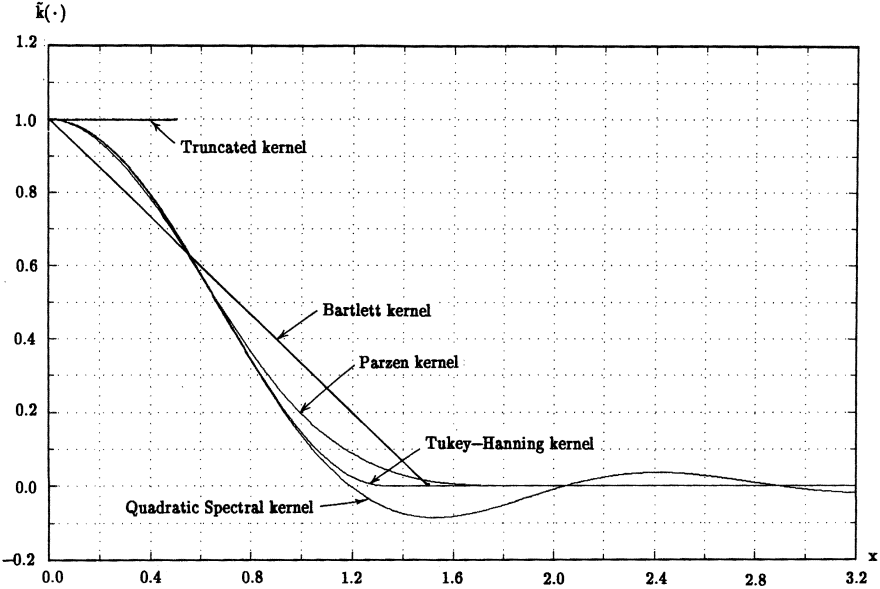

FIGURE 1.—Comparison of kernels.[a]

Source: Andrews (1991)

Kernel estimators: consistency

- Andrews (1991) is the authoritative analysis of kernel HAC.
  - Builds on classic stats literature on estimating spectral densities.
- Andrews shows that $\hat{\Omega} \xrightarrow{p} \Omega$ as $T \to \infty$ provided that:
  1. The kernel is cts at all but a finite number of points, cts at 0, and satisfies
  $$k(0) = 1, \quad k(-x) = k(x), \quad |k(x)| \leq 1, \quad \int_{-\infty}^{\infty} k(x)^2 \, dx < \infty.$$
  2. $\{y_t\}$ is weakly dependent and satisfies some other regularity conditions.
  3. $S_T \to \infty$ and $S_T/T \to 0$ as $T \to \infty$.

Kernel estimators: bias and variance

- To pick the kernel and the bandwidth, we need to analyze the bias-variance trade-off.

- Andrews (1991) also shows that, if $S_T \to \infty$ and $S_T/T \to 0$,

$$\mathrm{Var}(\mathrm{vec}(\hat{\Omega})) = \frac{S_T}{T} \int_{-\infty}^{\infty} k(x)^2 \, dx \times (I_{n^2} + K_{nn})(\Omega \otimes \Omega) + o(S_T/T).$$

  K_{nn} is the $n^2 \times n^2$ matrix s.t. $\mathrm{vec}(A') = K_{nn}\,\mathrm{vec}(A)$ for any $A \in \mathbb{R}^{n \times n}$.

- To state the bias, define first the modified q-th derivative of the kernel at 0,

$$k_q \equiv \lim_{x \to 0} \frac{1 - k(x)}{|x|^q}, \quad q \geq 0,$$

  and the modified q-th derivative of the spectrum at freq. 0,

$$s_y^{(q)}(0) \equiv \sum_{\ell=-\infty}^{\infty} |\ell|^q \Gamma_y(\ell), \quad q \geq 0.$$

Kernel estimators: bias and variance (cont.)

$$k_q \equiv \lim_{x \to 0} \frac{1 - k(x)}{|x|^q}, \quad s_y^{(q)}(0) \equiv \sum_{\ell=-\infty}^{\infty} |\ell|^q \Gamma_y(\ell)$$

- The **Parzen characteristic exponent** q^* of the kernel is given by the largest q such that $k_q < \infty$.
- Bartlett kernel: $q^* = 1$. Parzen and QS: $q^* = 2$. There exists no guaranteed positive semidefinite kernel estimator with $q^* > 2$.
- If $S_T \to \infty$, $S_T/T \to 0$, $S_T^{q^*}/T \to 0$, and $\|s_y^{(q^*)}(0)\| < \infty$, then

$$E(\hat{\Omega}) - \Omega = -S_T^{-q^*} k_{q^*} s_y^{(q^*)}(0) + o(S_T^{-q^*}).$$

- The bias of $\hat{\Omega}_{jj}$ is likely to be negative, since $k_{q^*} > 0$, and typically $s_{y_j}^{(q^*)}(0) > 0$ given positive serial correlation.

Kernel estimators: MSE

- Hence, if $S_T = o(T^{1/q^*})$, the variance is of order S_T/T and the bias is of order $S_T^{-q^*}$.
- Intuitively:
  - Variance increasing in S_T since more noisy ACF lags contribute to $\hat{\Omega}$.
  - Bias decreasing in S_T (at least for $S_T \ll T$) since we down-weight higher-lag ACFs less. Bias is larger when $s_y(\omega)$ changes rapidly for $\omega \approx 0$.
- The above results on bias and variance imply that, if $S_T = o(T^{1/q^*})$, the weighted mean square error (MSE) equals

$$
\begin{aligned}
MSE(S_T) &\equiv E[\operatorname{vec}(\hat{\Omega} - \Omega)' \mathcal{W} \operatorname{vec}(\hat{\Omega} - \Omega)] \\
&\approx \frac{S_T}{T} \int_{-\infty}^{\infty} k(x)^2 \, dx \times \operatorname{tr}\{\mathcal{W}(I_n + K_{nn})(\Omega \otimes \Omega)\} \\
&\quad + S_T^{-2q^*} k_{q^*}^2 \operatorname{vec}\left(s_y^{(q^*)}(0)\right)' \mathcal{W} \operatorname{vec}\left(s_y^{(q^*)}(0)\right).
\end{aligned}
$$

Kernel estimators: MSE-optimal bandwidth

- The approximate MSE has a unique minimum in S_T:

$$S_T^* \equiv \left(\frac{q^* k_{q^*}^2 \alpha(q^*)}{\int_{-\infty}^{\infty} k(x)^2 \, dx} T \right)^{1/(2q^*+1)},$$

$$\alpha(q^*) \equiv \frac{2 \operatorname{vec}\left(s_y^{(q^*)}(0)\right)' \mathcal{W} \operatorname{vec}\left(s_y^{(q^*)}(0)\right)}{\operatorname{tr}\{\mathcal{W}(I_n + K_{nn})(\Omega \otimes \Omega)\}}.$$

- $S_T^* \propto T^{1/(2q^*+1)}$ balances variance and *squared* bias. The optimal MSE is thus of order $MSE(S_T^*) = O(T^{-2q^*/(2q^*+1)})$.

- Not surprisingly, the optimal bandwidth depends on the true spectrum near 0. Oft-used options:

  1. Andrews (1991): Estimate Ω and $s_y^{(q^*)}(0)$ using a preliminary VAR(1) model; then plug into S_T^* formula to obtain final $\hat{\Omega}$.

  2. MSE-optimal bandwidth for Newey-West given AR(1) with $\rho = 0.25$: $S_T^* = 0.75\, T^{1/3}$.

Kernel estimators: MSE-optimal kernel

- Since the MSE, given optimal bandwidth, is of order $T^{-2q^*/(2q^*+1)}$, it is certainly optimal to choose a kernel with $q^* = 2$ (recall: no kernel with $q^* > 2$ guarantees a p.s.d. $\hat{\Omega}$). So Bartlett is dominated.
- Which kernel with $q^* = 2$ minimizes the MSE?
- We normalize $\int_{-\infty}^{\infty} k(x)^2 \, dx = 1$. This makes the bandwidth comparable across kernels. E.g., given any kernel $k(\cdot)$, the kernel $\tilde{k}(x/c) = k(x)$ is like changing the bandwidth S_T of $k(\cdot)$ to $c \times S_T$.
- $MSE(S_T)$ then only depends on the kernel through

$$k_2^2 = \left(\lim_{x \to 0} \frac{1 - k(x)}{x^2} \right)^2 = \frac{1}{4} (k''(0))^2,$$

  which we want to minimize subject to $\int_{-\infty}^{\infty} k(x)^2 \, dx = 1$ and $k(\cdot)$ guaranteeing a p.s.d. $\hat{\Omega}$
- This calculus of variations problem is solved by the QS kernel.

Kernel estimators: taking stock

- We have a family of kernel estimators that are consistent.
- We have a method for estimating the optimal bandwidth, given a choice of kernel.
- The MSE, given optimal bandwidth, is minimized by the QS kernel.
- The Bartlett kernel (Newey-West) leads to a slower MSE convergence rate ($T^{-2/3}$) than the QS kernel ($T^{-4/5}$).
- In practice, however, results are often seen to be much more sensitive to the bandwidth than the kernel.
- Monte Carlo studies have found that confidence intervals based on MSE-optimal bandwidth/kernel can drastically undercover, if the underlying data is moderately or highly persistent (more soon).

Kernel estimators: why poor finite-sample performance?

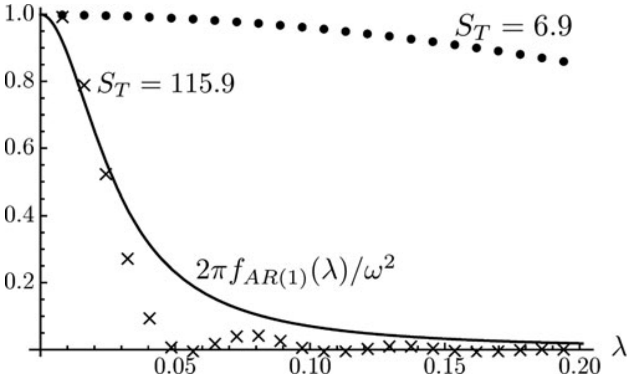

Figure 2. Newey–West weights on periodogram ordinates and normalized AR(1) spectral density. Notes: The dots and crosses correspond to the approximate weights on the first 24 periodogram ordinates of a Newey–West LRV estimator with bandwidth equal to S_T, normalized by the weight on the first periodogram ordinate. Total weight $\sum_{l=1}^{(T-1)/2} K_{T,l}$ is 0.99 and 0.85 for $S_T = 6.9$ and 115.9, respectively. The line is the spectral density of an AR(1) process with coefficient 0.973, scaled by $2\pi/\omega^2$. Source: Müller (2014)

Kernel estimators: prewhitening

- Andrews & Monahan (1992), following the statistics literature, found that **prewhitening** can somewhat improve the performance of kernel HAC estimators.
- Idea: First fit a VAR(p) model to the data (typically $p = 1$). Then use a kernel HAC estimator for the LRV of the VAR residuals. Compute the implied LRV of the data itself:

$$\hat{\Omega} = (I_n - \textstyle\sum_{\ell=1}^{p} \hat{A}_\ell)^{-1} \tilde{\Omega} (I_n - \textstyle\sum_{\ell=1}^{p} \hat{A}_\ell)^{-1\prime},$$

  where $\tilde{\Omega}$ is a kernel HAC estimator applied to the VAR *residuals*.
- The preliminary VAR step removes most of the dependence in the data, so the kernel HAC estimator may work OK on the near-serially-uncorrelated residuals.

Kernel estimators: impossibility results

- Pötscher (2002) and Müller (2007) give different arguments for why *uniformly* consistent LRV estimation is impossible.
- Intuitively, if an estimator $\hat{\Omega}$ is consistent for the true LRV for a particular DGP, it can be "fooled" by a slightly different DGP where the spectrum is the same at all frequencies except very close to 0.
- Ultimately, we can only be confident that an estimator $\hat{\Omega}$ is accurate for a given sample size if we are willing to make *a priori* assumptions about the shape of the spectrum near 0 (i.e., about the persistence properties of the data).
- Dou (2024) performs optimal inference under such explicit smoothness conditions on the spectrum near 0.

Long-run variance as an ingredient in testing

- So far we have tried to minimize the MSE of $\hat{\Omega}$.
- But typically, we don't care about Ω *per se*, it's just an ingredient to constructing hypothesis tests or confidence intervals.
- MSE-minimization may not generate optimal properties of tests/CIs.

Kernel estimators: testing-optimal bandwidth

- Sun, Phillips & Jin (2008) and Sun (2014) derive the bandwidth S_T that delivers highest power of a HAC-based test, given that the size of the test is close to α.

- It turns out that in this calculation, we should give more weight to bias than a MSE calculation indicates. Instead of balancing *squared* bias and variance, we should just balance bias and variance. See the slide appendix for a derivation sketch.

- This leads us to consider a larger bandwidth S_T. Specifically, in most cases the optimal bandwidth is

$$S_T^* \propto T^{1/(1+q^*)},$$

  where q^* is the Parzen characteristic exponent of the kernel.

- For QS, $S_T^* \propto T^{1/3}$. For Bartlett, $S_T^* \propto T^{1/2}$.

Kernel estimators: fixed-b asymptotics

- Our standard asymptotics for GMM, say, only use the fact $\hat{\Omega} \xrightarrow{p} \Omega$. E.g., if $\{y_t\}$ is univariate and $\Omega \equiv \text{LRV}(y_t)$,

$$\sqrt{T}\hat{\Omega}^{-1/2}(\hat{\mu}_y - \mu_y) \xrightarrow{d} \Omega^{-1/2} N(0, \Omega) = N(0, 1).$$

- Neither the kernel nor the bandwidth show up in the limiting distribution. Effectively, the usual asymptotics are *as if* Ω were known, whereas it actually is difficult to estimate in practice.

- Kiefer & Vogelsang (2002, 2005) propose an alternative asymptotic approximation with better small-sample accuracy.

Kernel estimators: fixed-b asymptotics (cont.)

- Define $b_T \equiv S_T/T$. $\hat{\Omega}$ denotes a kernel HAC estimator.
- The Andrews (1991) asymptotics assume $b_T \to 0$, which yields consistency of $\hat{\Omega}$.
- The **fixed-b asymptotics** of Kiefer & Vogelsang instead assume that b_T is fixed at a constant $b \in (0, 1]$ as $T \to \infty$. Then $\hat{\Omega}$ is no longer consistent but converges in distribution to a nondegenerate limit as $T \to \infty$.
    - We should not think of the fixed-b framework as a different estimator.
    - It is just a different approximation to the finite-sample sampling distribution of $\hat{\Omega}$: We do not assume that b_T is negligibly small, so the randomness of $\hat{\Omega}$ matters asymptotically for t- or F-tests (as it does in finite samples).

Kernel estimators: fixed-b asymptotics (cont.)

- Kiefer & Vogelsang (2005) derive the limiting distribution of GMM test statistics under fixed-b asymptotics.

- Under various regularity conditions, the usual GMM t-statistic for the null hypothesis $H_0\colon \lambda'\theta_0 = a$ has limiting null distribution

$$\frac{\lambda'\hat{\theta} - a}{\widehat{\text{se}}(\lambda'\hat{\theta})} \xrightarrow{d} \frac{N(0,1)}{\Xi_{k(\cdot),b}}.$$

- Here $\Xi_{k(\cdot),b}$ is a positive random variable that depends only on the kernel $k(\cdot)$ and the bandwidth constant b. The numerator and denominator are independent.

- The proof uses a technical tool called the Functional Central Limit Theorem. Will return to this later in the course.

Kernel estimators: fixed-b asymptotics (cont.)

- The t-test thus remains *asymptotically pivotal* under the null, like under the standard asymptotics.
- This means that it's easy to simulate critical values based on an auxiliary artificial model:

1. Generate fake $\tilde{y}_t \overset{iid}{\sim} N(0,1)$ for $t = 1, \ldots, \tilde{T}$, with $\tilde{T}$ very large.
2. Use HAC estimator $\tilde{\Omega}$ to estimate $\text{LRV}(\tilde{y}_t)$, with the *same* kernel and b parameter as used in the actual GMM problem (so in the fake data, the HAC bandwidth equals $b\tilde{T}$).
3. Construct t-statistic $\tilde{t} \equiv \tilde{\Omega}^{-1/2}\tilde{T}^{-1/2}\sum_{t=1}^{\tilde{T}} \tilde{y}_t$ of $H_0\colon E(\tilde{y}_t) = 0$ (which is true in the fake data).
4. Repeat 1–3 many times. Compute crit. val. as the $1-\alpha$ quantile of $|\tilde{t}|$.

- For joint testing, we can do the same for the F-statistic.

Kernel estimators: fixed-b asymptotics (cont.)

- The nonstandard limiting distributions under fixed-b asymptotics converge to the usual $N(0, 1)$ or χ^2 distributions as $b \to 0$. That is, the usual asymptotics are a limiting case.
- Why nonstandard distribution? Because the denominators in the test statistics (which depend on $\hat{\Omega}$) remain random in the limit $T \to \infty$.
- To de-emphasize consistency of $\hat{\Omega}$, we talk of **heteroskedasticity and autocorrelation robust (HAR)** inference.
- Kiefer & Vogelsang find in simulations that the new critical values for t-tests and F-tests lead to substantially better size control.
- Jansson (2004) shows that (in a Gaussian location model) the fixed-b asymptotics are an **asymptotic refinement** of the normal asymptotics: The fixed-b asymptotics capture higher-order terms in the sampling distribution of HAR test statistics.

Optimal bandwidth choice under fixed-b asymptotics

- Lazarus, Lewis, Stock & Watson (2018) derive bandwidth rules that optimize the size-power trade-off (according to some reasonable loss function) for t-tests based on kernel HAR with fixed-b crit. val.
- For Bartlett (Newey-West) kernel, they recommend $S_T = 1.3T^{1/2}$.

Table 1. Monte Carlo rejection rates for 5% HAR t-tests under the null

	Estimator	Truncation rule	Critical values	Null imposed?	$\rho = 0.3$	$\rho = 0.5$	$\rho = 0.7$
1	NW	$S = 0.75T^{1/3}$	N(0,1)	No	0.088	0.114	0.180
2	NW	$S = 1.3T^{1/2}$	fixed-b (nonstandard)	No	0.067	0.079	0.108
3	EWC	$\nu = 0.4T^{2/3}$	fixed-b (t_ν)	No	0.062	0.071	0.097
4	NW	$S = 0.75T^{1/3}$	N(0,1)	Yes	0.077	0.095	0.149
5	NW	$S = 1.3T^{1/2}$	fixed-b (nonstandard)	Yes	0.056	0.060	0.074
6	EWC	$\nu = 0.4T^{2/3}$	fixed-b ($t\nu{<}\Sigma\psi\mu{>}$)	Yes	0.052	0.053	0.063
Theoretical bound based on Edgeworth expansions for the Gaussian location model							
7	NW	$S = 1.3T^{1/2}$	fixed-b (nonstandard)	No	0.054	0.058	0.067
8	EWC	$\nu = 0.4T^{2/3}$	fixed-b ($t\nu{<}\Sigma\psi\mu{>}$)	N/A	0.051	0.054	0.064

NOTES: Tests are on the single stochastic regressor in (1) with nominal level 5%, $T = 200$. The single stochastic regressor x_t and disturbance u_t are independent Gaussian AR(1)'s, with AR coefficients $\rho_X = \rho_u = \rho^{1/2}$, where ρ is given in the column heading. All regressions include an intercept. The tests differ in the kernel used for the standard errors (first column; NW is Newey-West/Bartlett kernel, EWC is equal-weighted cosine transform). The truncation rule for NW is expressed in terms of the truncation parameter S and for EWC as the degrees of freedom ν of the test, which is the number of cosine terms included. The third and fourth columns describe the critical values used and whether the estimator of Ω is computed under the null or is unrestricted. For the EWC test in the location model, the test statistic is the same whether the null is imposed or not.

Source: Lazarus, Lewis, Stock & Watson (2018)

Equal-weighted cosine estimator

- The LRV is about what happens at very low frequencies.
- Idea: Project data $\{y_t\}$ onto a small number B of deterministic basis functions that capture low-frequency cycles, such as cosines $\cos\left(\omega_\ell(t - \frac{1}{2})\right)$ with frequency $\omega_\ell = \pi\ell/T$ close to 0, $\ell = 1, \dots, B$.
- Then set $\hat{\Omega} \equiv ESS$ (explained sum of squares) from this projection.
- For fixed B, $\hat{\Omega}$ will be inconsistent, but can analyze its asy. distr.
- L, L, S & W (2018) find through theory and simulations that this **equal-weighted cosine (EWC) estimator** works well. Turns out that we can get crit. val's for t-test based on $\hat{\Omega}$ from the standard Student-t distribution. Simple and effective!
- Related references: Phillips (2005), Müller (2007), Sun (2013).

Appendix: Bias-variance trade-off in testing

- Following Stock & Watson's 2008 NBER lectures, suppose for simplicity that an estimator $\hat{\mu}$ is distributed $\hat{\mu} \sim N(0, \Omega)$, and we have available an estimator $\hat{\Omega}$ that is independent of $\hat{\mu}$.
- We want to approx. the sampling distribution of the t-statistic $\frac{\hat{\mu}}{\hat{\Omega}^{1/2}}$.
- Let $F(\cdot)$ be the CDF of a χ^2_1 variable. For any $c \in \mathbb{R}$,

$$\begin{aligned} P(\tfrac{\hat{\mu}^2}{\hat{\Omega}} \leq c) &= E[P(\tfrac{\hat{\mu}^2}{\Omega} \leq c\tfrac{\hat{\Omega}}{\Omega} \mid \hat{\Omega})] \\ &= E[F(c\tfrac{\hat{\Omega}}{\Omega})] \\ &\approx F(c) + cF'(c)\underbrace{E[\tfrac{\hat{\Omega}-\Omega}{\Omega}]}_{\text{bias}} + \tfrac{1}{2}c^2F''(c)\underbrace{E[\tfrac{(\hat{\Omega}-\Omega)^2}{\Omega^2}]}_{\text{MSE=bias}^2\text{+var.}}. \end{aligned}$$

- If both the bias and variance tend to zero as $T \to \infty$, then bias is larger than bias^2. Hence, the dominant terms to understand are the bias and variance, not the bias squared and the variance.